

MAT110

Differential Calculus and Coordinate Geometry

Lecture 12

Clairaut's Theorem and Taylor Series for Two Variables

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1 Clairaut's Theorem

Clairaut's Theorem

Suppose that f is defined on a disk D that contains the point (a, b) . If the functions f_{xy} and f_{yx} are continuous on the disk, then

$$f_{xy}(a, b) = f_{yx}(a, b).$$

Similarly, when the required higher-order partial derivatives are continuous, their order can be interchanged. For example,

$$f_{xyx} = f_{yxx}.$$

For the second-order derivatives of $f(x, y)$,

$$f_{xy} = f_{yx},$$

so the distinct terms are f_{xx} , $2f_{xy}$, and f_{yy} . For the third-order derivatives,

$$f_{xxy} = f_{xyx} = f_{yxx}, \quad f_{xyy} = f_{yxy} = f_{yyx},$$

so the corresponding terms are

$$f_{xxx}, \quad 3f_{xxy}, \quad 3f_{xyy}, \quad f_{yyy}.$$

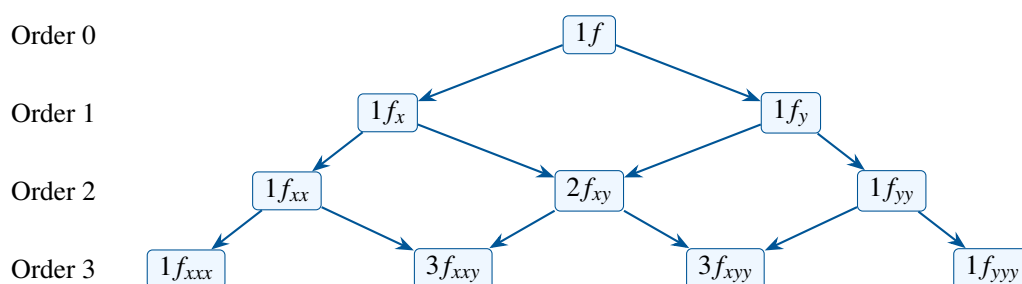
Let $f(s, r, t)$ be a function of s , r , and t . According to Clairaut's theorem,

$$f_{ssrtsrr} = f_{trsrssr},$$

because in each case the differentiation is performed with respect to t once, s three times, and r three times.

2 Derivative Tree for $f(x, y)$

The coefficient pattern needed for the Taylor expansion is displayed in the following tree.



3 Taylor Series Expansion for Two Variables

The Taylor series of a real or complex function $f(x)$ that is infinitely differentiable in a neighborhood of a real or complex number a is the power series

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \frac{f^{(4)}(a)}{4!}(x-a)^4 + \dots$$

Taylor Series of $f(x,y)$ at (a,b)

The Taylor series for a function $f(x,y)$ at (a,b) is

$$\begin{aligned} f(x,y) = & f(a,b) + \frac{f_x(a,b)}{1!0!}(x-a)(y-b)^0 + \frac{f_y(a,b)}{0!1!}(x-a)^0(y-b) \\ & + \frac{f_{xx}(a,b)}{2!}(x-a)^2 + \frac{2f_{xy}(a,b)}{2!}(x-a)(y-b) + \frac{f_{yy}(a,b)}{2!}(y-b)^2 \\ & + \frac{f_{xxx}(a,b)}{3!}(x-a)^3 + \frac{3f_{xxy}(a,b)}{3!}(x-a)^2(y-b) \\ & + \frac{3f_{xyy}(a,b)}{3!}(x-a)(y-b)^2 + \frac{f_{yyy}(a,b)}{3!}(y-b)^3 + \dots \end{aligned}$$

Equivalently,

$$\begin{aligned} f(x,y) = & f(a,b) + \frac{1}{1!} \left[f_x(a,b)(x-a) + f_y(a,b)(y-b) \right] \\ & + \frac{1}{2!} \left[f_{xx}(a,b)(x-a)^2 + 2f_{xy}(a,b)(x-a)(y-b) + f_{yy}(a,b)(y-b)^2 \right] \\ & + \frac{1}{3!} \left[f_{xxx}(a,b)(x-a)^3 + 3f_{xxy}(a,b)(x-a)^2(y-b) \right. \\ & \quad \left. + 3f_{xyy}(a,b)(x-a)(y-b)^2 + f_{yyy}(a,b)(y-b)^3 \right] + \dots \end{aligned}$$

Important Note

When you are asked to calculate a Taylor polynomial, do not include the dots ($\dots$). The dots are used to indicate the continuing series expansion.

4 Taylor Polynomials

4.1 First-Degree Taylor Polynomial

The first-degree Taylor polynomial of $f(x,y)$ near (a,b) is

$$P_1(x,y) = L(x,y) = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b).$$

Here $L(x,y)$ is called the linear (or tangent-plane) approximation of f for (x,y) near (a,b) .

4.2 Second-Degree Taylor Polynomial

The second-degree Taylor polynomial is

$$\begin{aligned} P_2(x,y) = Q(x,y) = & P_1(x,y) + \frac{1}{2!} \left[f_{xx}(a,b)(x-a)^2 \right. \\ & \left. + 2f_{xy}(a,b)(x-a)(y-b) + f_{yy}(a,b)(y-b)^2 \right]. \end{aligned}$$

4.3 Third-Degree Taylor Polynomial

The third-degree Taylor polynomial is

$$P_3(x, y) = C(x, y) = Q(x, y) + \frac{1}{3!} \left[f_{xxx}(a, b)(x-a)^3 + 3f_{xxy}(a, b)(x-a)^2(y-b) + 3f_{xyy}(a, b)(x-a)(y-b)^2 + f_{yyy}(a, b)(y-b)^3 \right].$$

5 Examples

Example 01

Calculate the second-degree Taylor polynomial of

$$f(x, y) = \sin 2x + \cos y$$

near $(0, 0)$.

Solution

Given

$$f(x, y) = \sin 2x + \cos y, \quad f(0, 0) = 1.$$

The first-order partial derivatives are

$$\begin{aligned} f_x(x, y) &= 2 \cos 2x, & f_x(0, 0) &= 2, \\ f_y(x, y) &= -\sin y, & f_y(0, 0) &= 0. \end{aligned}$$

The second-order partial derivatives are

$$\begin{aligned} f_{xx}(x, y) &= -4 \sin 2x, & f_{xx}(0, 0) &= 0, \\ f_{xy}(x, y) &= 0 = f_{yx}(x, y), & f_{xy}(0, 0) &= 0, \\ f_{yy}(x, y) &= -\cos y, & f_{yy}(0, 0) &= -1. \end{aligned}$$

Therefore,

$$\begin{aligned} Q(x, y) &= 1 + \frac{1}{1!} [2(x-0) + 0(y-0)] \\ &\quad + \frac{1}{2!} [0(x-0)^2 + 0(x-0)(y-0) - 1(y-0)^2] \\ &= 1 + 2x - \frac{y^2}{2}. \end{aligned}$$

Thus,

$$f(x, y) \approx 1 + 2x - \frac{y^2}{2}.$$

Example 02

Expand

$$x^2y + 3y - 2$$

in powers of $(x - 1)$ and $(y + 2)$.**Solution**Use Taylor's expansion with $a = 1$ and $b = -2$, that is, near $(1, -2)$. Let

$$f(x, y) = x^2y + 3y - 2.$$

Then

$$f(1, -2) = -10.$$

The first-order partial derivatives are

$$\begin{aligned} f_x(x, y) &= 2xy, & f_x(1, -2) &= -4, \\ f_y(x, y) &= x^2 + 3, & f_y(1, -2) &= 4. \end{aligned}$$

The second-order partial derivatives are

$$\begin{aligned} f_{xx}(x, y) &= 2y, & f_{xx}(1, -2) &= -4, \\ f_{xy}(x, y) &= 2x, & f_{xy}(1, -2) &= 2, \\ f_{yy}(x, y) &= 0, & f_{yy}(1, -2) &= 0. \end{aligned}$$

The third-order partial derivatives are

$$\begin{aligned} f_{xxx} &= 0, & f_{xxx}(1, -2) &= 0, \\ f_{xxy} &= 2, & f_{xxy}(1, -2) &= 2, \\ f_{xyy} &= 0, & f_{xyy}(1, -2) &= 0, \\ f_{yyy} &= 0, & f_{yyy}(1, -2) &= 0. \end{aligned}$$

All higher derivatives are zero. Hence,

$$\begin{aligned} f(x, y) &= -10 - 4(x - 1) + 4(y + 2) \\ &\quad + \frac{1}{2!} \left[-4(x - 1)^2 + 2(2)(x - 1)(y + 2) + 0(y + 2)^2 \right] \\ &\quad + \frac{1}{3!} \left[0 + 3(2)(x - 1)^2(y + 2) + 0 + 0 \right] \\ &= -10 - 4(x - 1) + 4(y + 2) - 2(x - 1)^2 \\ &\quad + 2(x - 1)(y + 2) + (x - 1)^2(y + 2). \end{aligned}$$

Therefore,

$$f(x, y) = -10 - 4(x - 1) + 4(y + 2) - 2(x - 1)^2 + 2(x - 1)(y + 2) + (x - 1)^2(y + 2).$$

Problem

Expand the function

$$f(x,y) = x^3 + 3x^2y^2 + 4xy^2 + y^3$$

by Taylor's theorem in powers of $(x - 1)$ and $(y - 1)$.